Optimal Provable Robustness of Quantum Classification via Quantum Hypothesis Testing

Maurice Weber

Collaboration with:

Nana Liu, Bo Li, Ce Zhang, Zhikuan Zhao

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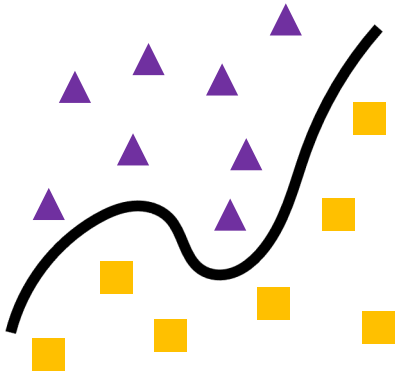
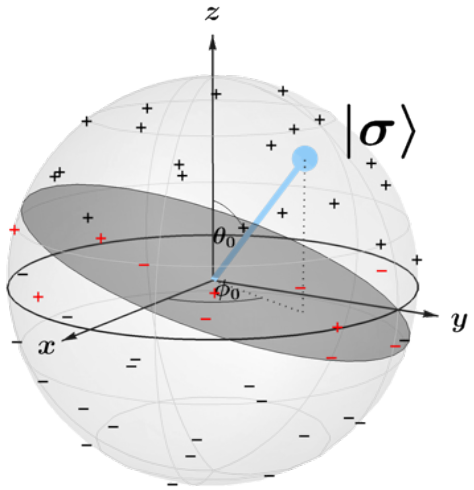
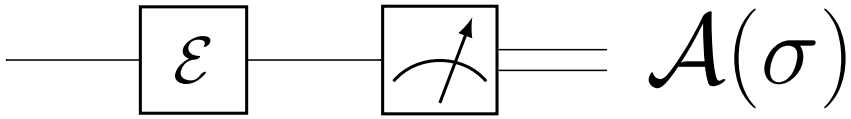
UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

自然科学研究院 Institute of Natural Sciences

Quantum Classifiers



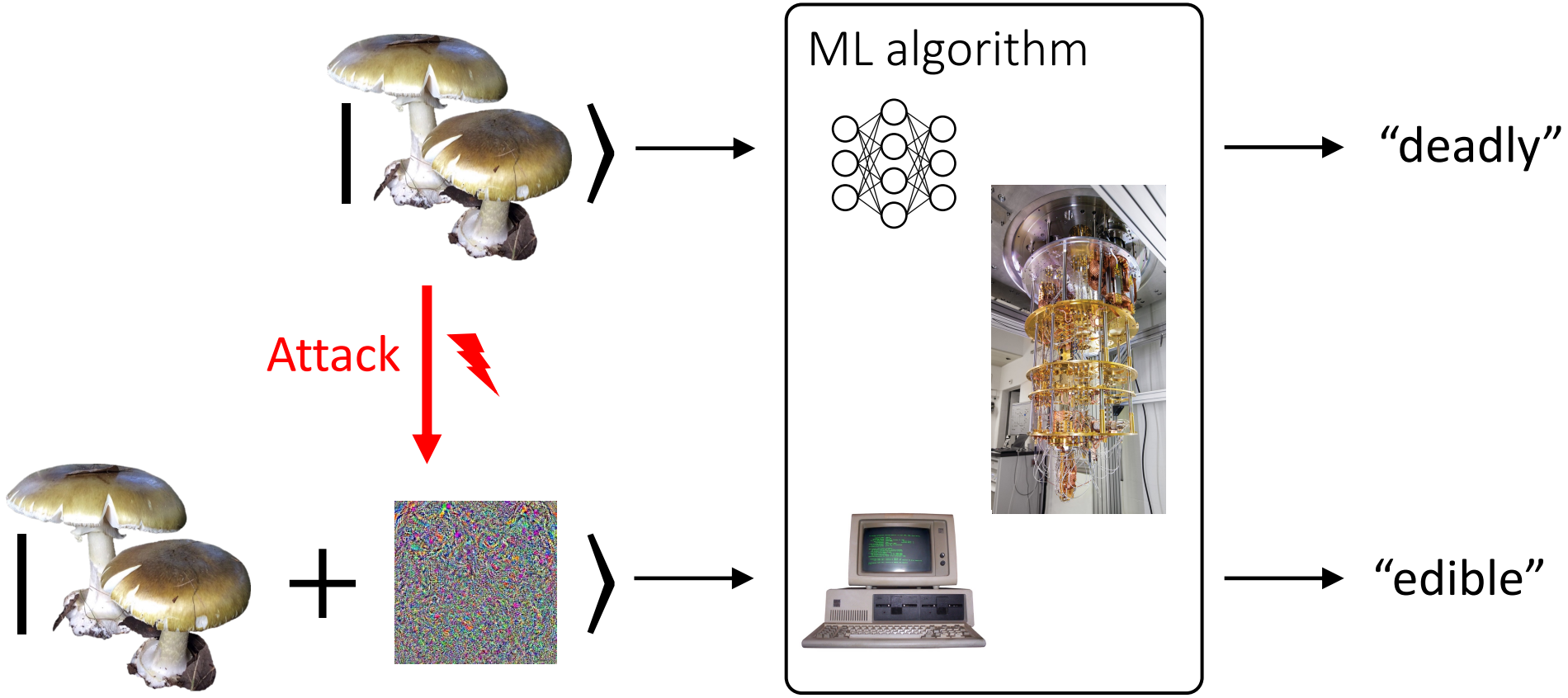




- Learn a rule to assign classes to datapoints, $\sigma \mapsto \mathcal{A}(\sigma) \in \{1, \ldots, K\}$.
- Such a classifier is described by a quantum channel $\mathcal{E}$ and a POVM $\{\Pi_k\}_k$.
- For a state σ, the probability of measuring outcome k is identified with the class probability

$$\mathcal{A}(\sigma) = \arg\max_k \mathbf{y}_k(\sigma), \qquad \mathbf{y}_k(\sigma) = \mathrm{Tr}\left[\Pi_k \mathcal{E}(\sigma)\right]$$

Vulnerability to Adversarial Attacks



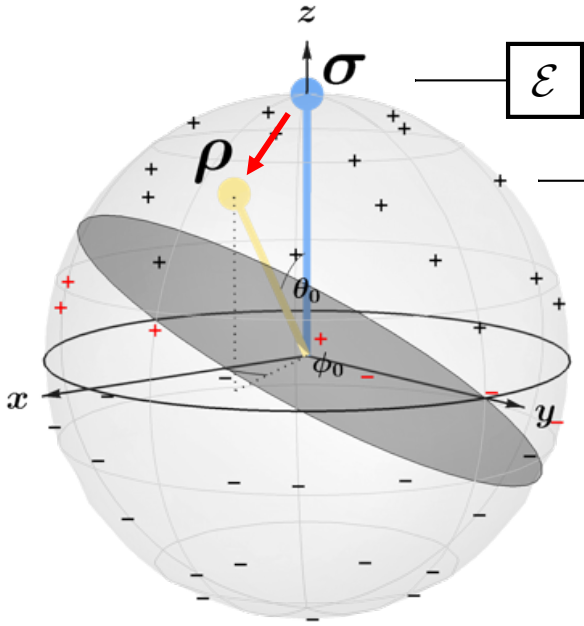
Q: How robust are quantum classifiers against these vulnerabilities?

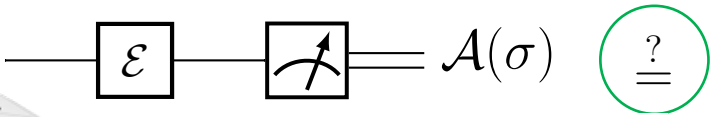
Robustness to Input Perturbations

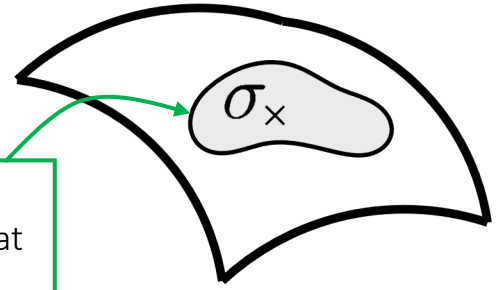
- Actual input state can be different from the intended one
- noisy implementations, adversaries
- Adversarial attack as worst-case noise
- ***Goal:*** characterize a *"robust region"* in which the prediction of our model does not change

noise, adversary

$$\sigma \longrightarrow \rho$$







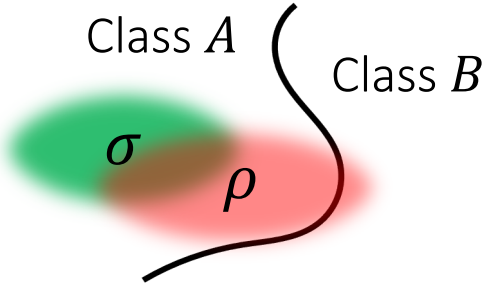
Robust region around σ where it is guaranteed that

$$\mathcal{A}(\sigma) = \mathcal{A}(\rho)$$

Robustness and Quantum Hypothesis Testing

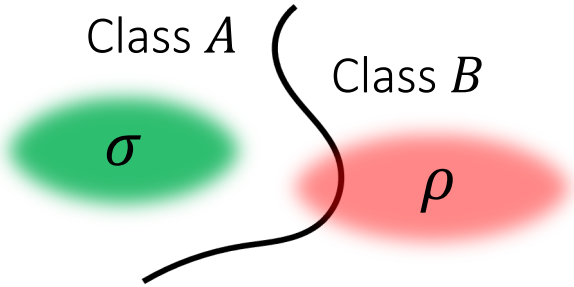
- Quantum classification and robustness:

  If the benign σ is in class A, when is the adversarial ρ also in class A?



- Quantum Hypothesis testing

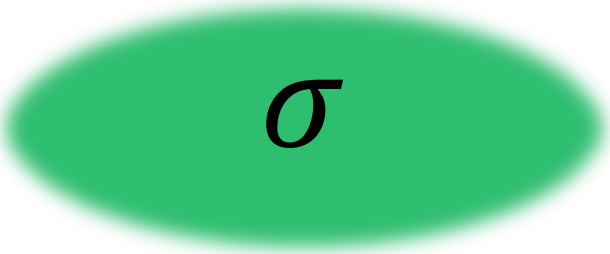
  Is the given state described by the benign σ or by the adversarial ρ?



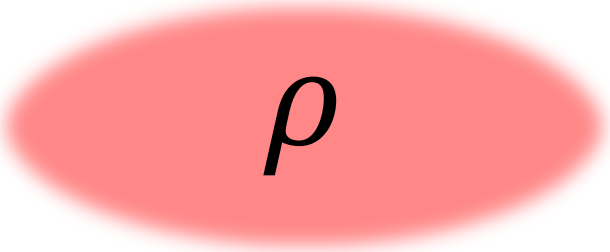
Quantum Hypothesis Testing (QHT)

Given a state which is described either by σ or ρ, the goal is to find a measurement $\{M, \mathbf{1} - M\}$ which optimally discriminates between the two hypotheses.

Null hypothesis described by σ



Alternative hypothesis described by ρ



Let $0 \leq M \leq \mathbb{1}$ be a hypothesis test

<u>*Type I error*</u>: σ tagged as alternative

$$\alpha(M;\, \sigma) = \mathrm{Tr}\,[M\sigma]$$

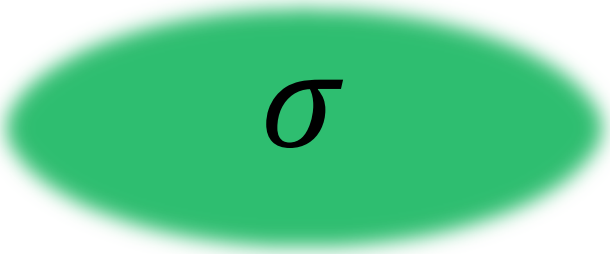
<u>*Type II error*</u>: ρ tagged as null

$$\beta(M;\, \rho) = \mathrm{Tr}\,[(\mathbb{1} - M)\rho]$$

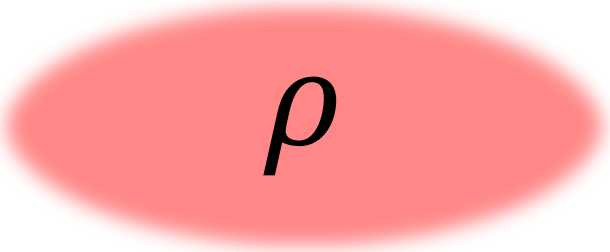
Quantum Hypothesis Testing (QHT)

Given a state which is described either by σ or ρ, the goal is to find a measurement $\{M, \mathbf{1} - M\}$ which optimally discriminates between the two.

Null hypothesis described by σ



Alternative hypothesis described by ρ



Type I error: σ tagged as alternative

$$\alpha(M;\,\sigma) = \mathrm{Tr}\,[M\sigma]$$

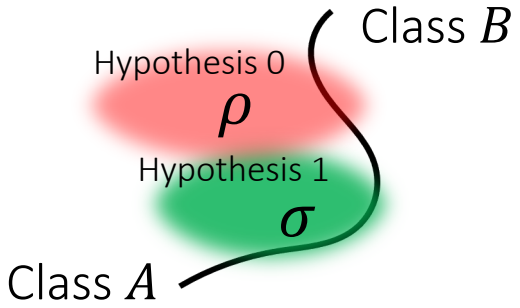
Type II error: ρ tagged as null

$$\beta(M;\,\rho) = \mathrm{Tr}\,[(\mathbb{1} - M)\rho]$$

Goal: $\beta^*(\alpha_0;\,\sigma,\,\rho) = \inf_{\alpha(M;\sigma)\leq\alpha_0} \beta(M;\,\rho)$

Robustness Guarantee via QHT

Intuition: *If the best possible discriminator cannot distinguish* σ *(null hypothesis) from* ρ *(alternative hypotheis), then the classifier is robust.*



$$\mathcal{A}(\sigma) = \arg\max_k \mathbf{y}_k(\sigma)$$

$$\mathbf{y}_k(\sigma) = \mathrm{Tr}\left[\Pi_k \mathcal{E}(\sigma)\right]$$

Theorem. Suppose that $A = \mathcal{A}(\sigma)$ *with class probabilities*

$$\mathbf{y}_A(\sigma) \geq p_A > p_B \geq \max_{k \neq A} \mathbf{y}_k(\sigma)$$

Then, for any ρ *which satisfies*

$$\beta^*(1 - p_A;\, \sigma,\, \rho) + \beta^*(p_B;\, \sigma,\, \rho) > 1$$

it is guaranteed that $\mathbf{y}_A(\rho) > \max_{k \neq A} \mathbf{y}_k(\rho)$ *, i.e.* $A = \mathcal{A}(\rho)$ *.*

QHT Robustness Bound

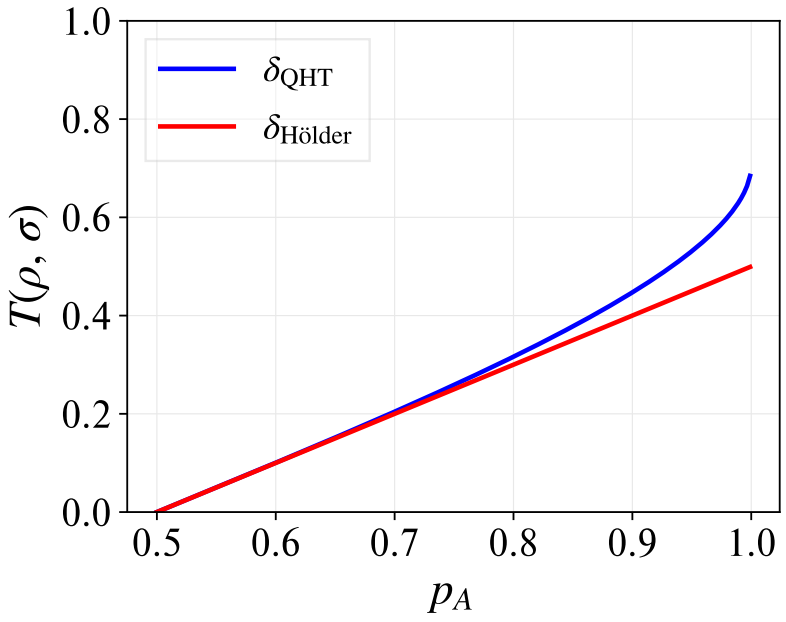
*<u>Corollary.</u> Let σ be a **pure state** with $A = \mathcal{A}(\sigma)$ with class probabilities satisfying $p_A + p_B = 1$.*

*Then, for any **pure state** ρ which satisfies*

$$T(\rho,\, \sigma) = \frac{1}{2}\,\|\rho - \sigma\|_1 < \sqrt{\frac{1}{2} - \sqrt{p_A(1-p_A)}} \equiv \delta_{\mathrm{QHT}}$$

it is guaranteed that $\mathbf{y}_A(\rho) > \max_{k \neq A} \mathbf{y}_k(\rho)$, i.e.

$$A = \mathcal{A}(\rho)$$



If σ and ρ are allowed to be **mixed**, then a sufficient condition is

$$T(\rho,\, \sigma) < \frac{p_A - p_B}{2} \equiv \delta_{\mathrm{Hölder}}$$

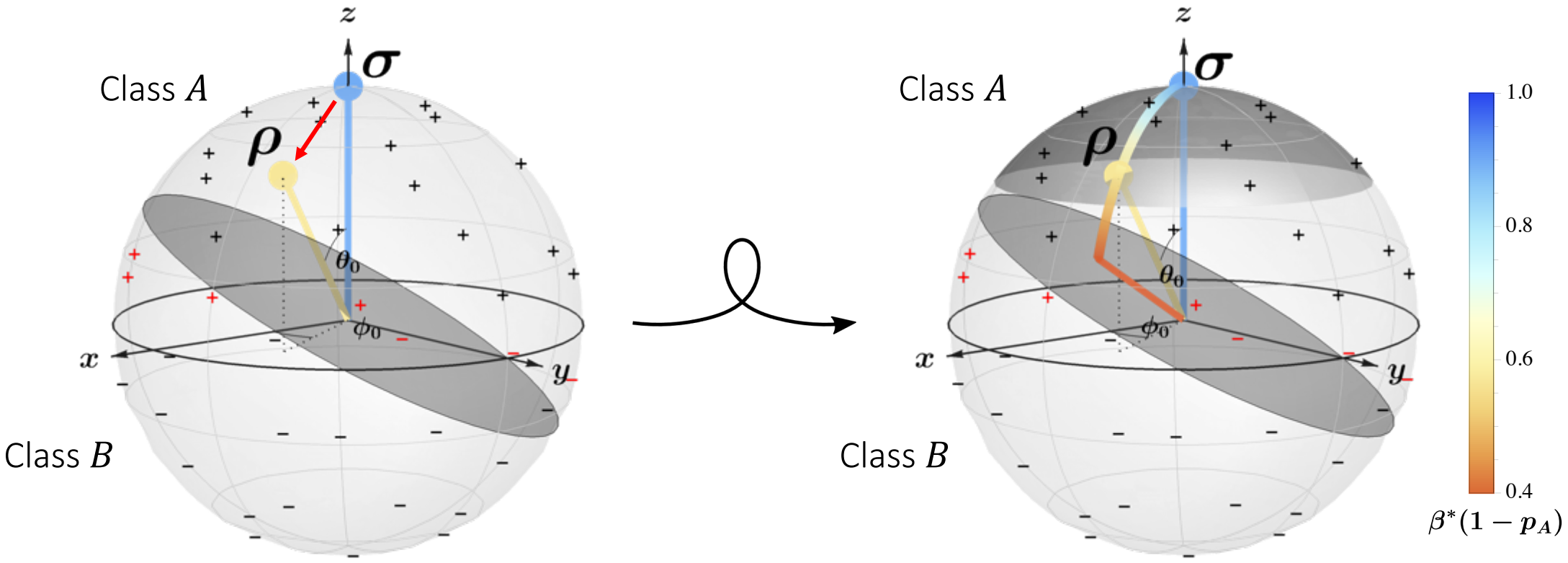
Tightness

Our robustness condition is tight in the following sense:

<u>Theorem.</u> In the case where $p_A + p_B = 1$ *, whenever the adversarial state* ρ *violates the robustness condition, then there exists a classifier* $\mathcal{A}^*$ *for which*

$$\mathcal{A}^*(\rho) \neq \mathcal{A}^*(\sigma)$$

QHT Robustness Bound: Example



$$p_A = \mathbf{y}_A(\sigma) = 0.9 \quad \Rightarrow \quad \begin{cases} \beta^*(1-p_A) = \beta^*(0.1) \approx 0.6 > 0.5 \\ T(\rho,\, \sigma) \approx 0.38 < \delta_{pure} \approx 0.44 \end{cases}$$

Conclusions

- We have seen how quantum hypothesis testing can be leveraged to obtain **robustness guarantees** for classification algorithms
- This robustness guarantee is **provably optimal** under modest assumptions
- We see this result as a first step to rigorously assessing reliability of quantum classifiers in noisy or adversarial settings
- Based on these results, we hope to inspire the future development of protocols that take into account adversarial robustness and investigate the trade off between robustness and the expressive power of QML.

Thank you!